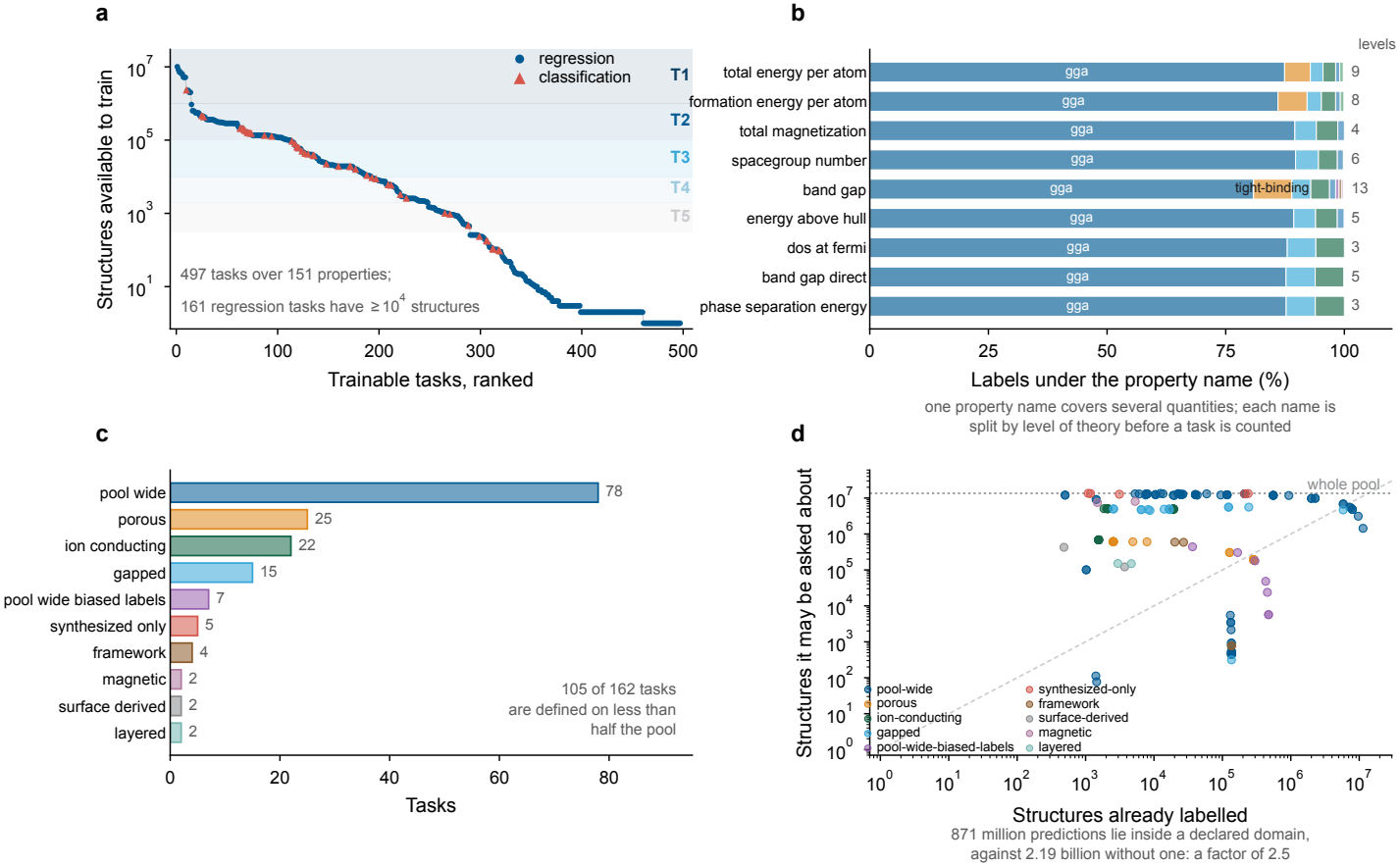
a
Structures available to train
regression
classification
T1
T2
T3
T4
T5
497 tasks over 151 properties;
161 regression tasks have ≥ 10^4 structures
Trainable tasks, ranked
b
levels
total energy per atom
formation energy per atom
total magnetization
spacegroup number
band gap
energy above hull
dos at fermi
band gap direct
phase separation energy
gga
tight-binding
Labels under the property name (%)
one property name covers several quantities; each name is split by level of theory before a task is counted
c
pool wide
porous
ion conducting
gapped
pool wide biased labels
synthesized only
framework
magnetic
surface derived
layered
105 of 162 tasks are defined on less than half the pool
Tasks
d
Structures it may be asked about
whole pool
pool-wide
porous
ion-conducting
gapped
pool-wide-biased-labels
synthesized-only
framework
surface-derived
magnetic
layered
Structures already labelled
871 million predictions lie inside a declared domain, against 2.19 billion without one: a factor of 2.5